

Beyond PSNR: Reward Design for NDVI-Consistent Super-Resolution of Sentinel-2 Imagery

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Section: 01, ID: 22301714

Abstract

Super-resolution (SR) models for satellite imagery are almost universally trained and evaluated with pixel-fidelity metrics such as PSNR and SSIM yet the products that downstream users actually consume are derived indices such as the Normalized Difference Vegetation Index (NDVI); whether the model that wins on PSNR is also the model that best preserves NDVI is rarely tested directly and reward functions are built purely around pixel fidelity which may therefore be optimizing the wrong target for scientific end-use. Here, this study builds a patch-level evaluation pipeline on the cross-sensor SEN2VEN μ S dataset (Sentinel-2 10 m $\rightarrow$ VEN μ S 5 m, $\times 2$ SR) that reports PSNR, SSIM, NDVI mean absolute error (MAE) and NDVI bias for every method through one shared evaluation routine, first establishing on 894 held-out test patches (five acquisition dates) that PSNR and NDVI error decorrelate (Spearman $\rho \approx -0.63$, $R^2 \leq 0.363$, and outright disagreement on the better of two methods on 15.5% of patches) and then testing via a minimal reinforcement-learning ablation of two tabular Q-learning agents differing only in whether their per-pixel reward includes an NDVI term, whether a reward that is told about NDVI can recover that loss. The NDVI-aware agent measurably shifts its policy toward the objective it is given (39.2% of learned state-action pairs diverge from the pixel-only agent, concentrated in the near-infrared-only actions built for exactly this purpose and it attains the smallest absolute NDVI bias of all four evaluated methods) but pays a PSNR cost without a matching NDVI MAE gain and neither Q-learning agent beats plain bicubic upsampling in or out-of-sample; a headroom analysis attributes this to the cross-sensor nature of the dataset itself an oracle given the held-out date’s own mean error recovers under 1% of the bicubic residual; rather than to a defect in the learning method.

1 Introduction

Vegetation indices computed from multispectral satellite imagery, most prominently NDVI [1], underpin a large share of operational remote sensing: crop monitoring, drought and vegetation-stress detection, and land-cover change analysis all consume NDVI rather than raw reflectance. The Sentinel-2 mission provides this imagery globally at 10m resolution, which is frequently too coarse for field-scale or sub-field-scale analysis, motivating single-image super-resolution as a way to synthesize sharper imagery without a new sensor. However, the overwhelming majority of the super-resolution literature is trained and evaluated with pixel-fidelity losses PSNR and SSIM, that are designed to correlate with perceived photographic quality on natural images, not with whether a physically meaningful derived quantity such as NDVI is preserved. If a model that scores well on PSNR is not necessarily the model that best preserves NDVI, then selecting, tuning or deploying an SR model purely by pixel fidelity risks silently degrading exactly the product that downstream science depends on and no further improvement in PSNR would ever reveal that this has happened. This project asks whether that risk is real at the pixel level and if it is, whether telling a learning agent about NDVI directly by putting it in the reward function of a reinforcement-learning (RL) agent rather than changing the architecture can do anything to close the gap.

The literature offers pieces of this puzzle but not the whole picture. SRCNN [2] established the now-standard end-to-end CNN mapping from a low-resolution (LR) to a high-resolution (HR) image, trained to minimize pixel error; it is effective at recovering pixel fidelity but is known to over-smooth high-frequency detail and it is not evaluated against any physically derived index. ESRGAN [3] demonstrated that optimizing for perceptual realism via an adversarial and perceptual loss systematically sacrifices PSNR relative to a pixel-loss network direct evidence that "looks better" and "is more pixel-accurate" are already known

to diverge, though only along a perceptual, human-facing axis. Blau and Michaeli [4] formalized this as the perception-distortion tradeoff, proving that any restoration method that improves perceptual quality must, beyond some point, sacrifice distortion metrics such as PSNR; a theoretical argument that no single fidelity number can certify a restoration model along every axis a user might care about, of which NDVI accuracy is one instance the theorem does not itself consider. On remote sensing specifically, Lanaras et al. [5] trained a CNN to super-resolve Sentinel-2’s 20 m and 60 m bands using its own 10 m bands as a supervisory signal, one of the few works to super-resolve Sentinel-2 multispectral data end to end but it too is evaluated with band-wise pixel error rather than any vegetation index; a recent survey of the field [6] confirms that PSNR/SSIM remain the dominant reported metrics across remote-sensing SR work, with downstream-task-aware evaluation still rare. On the reinforcement-learning side, RL-Restore [7] and Exposure [8] both show that treating image restoration or photo enhancement as a sequential decision process where an agent choosing discrete, interpretable edits region-by-region or step-by-step is a workable and interpretable alternative to a single end-to-end regression network, but both are still trained against pixel or perceptual objectives, never a downstream physical quantity. The tabular Q-learning [9] and the general reinforcement-learning formalism this project reuses directly are due to Sutton and Barto [10]. Finally, the SEN2VEN μ S dataset [11] is what makes a genuinely cross-sensor LR/HR benchmark possible at all, as opposed to the synthetically downsampled pairs (blur-then-downsample a single image) used by most of the SR literature above, this is a distinction that this project’s own headroom analysis (Section 5.1) shows is not cosmetic.

None of this prior work directly measures whether PSNR-optimized super-resolution degrades a specific, operationally important downstream product at the pixel level on real, non-synthetic cross-sensor pairs, nor whether simply reshaping the reward signal; leaving the model and architecture untouched can recover any of that loss. This project fills that specific, narrow gap with a small, controlled ablation rather than a new architecture: the same action space and the same tabular learner are used for both reward variants, so any difference in outcome is attributable to the reward alone.

1.1 Our Contribution

- A patch-level, acquisition-pair-disjoint evaluation pipeline on SEN2VEN μ S that reports PSNR, SSIM, NDVI MAE and NDVI bias for every method through one shared `evaluate()` routine, so every number in this report is directly comparable across methods.
- Direct, statistically tested evidence that PSNR and NDVI error decorrelate at the pixel level on real Sentinel-2/VEN μ S pairs: a correlation analysis, a within-patch paired disagreement test and a curated gallery of high-PSNR/high-NDVI-error failure cases.
- A minimal reward-shaping ablation: two tabular Q-learning agents, identical in seed, architecture and action space, differing only in whether the reward includes an NDVI term that isolates the causal effect of reward composition on learned policy.
- A causally diagnosed, honestly reported negative result for the RL component, supported by an in-sample check that rules out overfitting and a held-out oracle headroom analysis that rules out a fixable training defect, rather than a hidden or rationalized failure.

2 Material and Methods

Figure 1 summarizes the pipeline. A SEN2VEN μ S acquisition pair supplies a 10 m Sentinel-2 patch and its co-registered 5 m VEN μ S patch. Every method is bicubic upsampling, a trained SRCNN and two Q-learning agents starts from the same bicubically-upsampled image and is scored by the identical `evaluate()` routine, which computes PSNR, SSIM, NDVI MAE and NDVI bias against the true HR patch. Because every method is evaluated through the same function on the same held-out patches, the resulting four-way comparison and all significance tests are directly comparable.

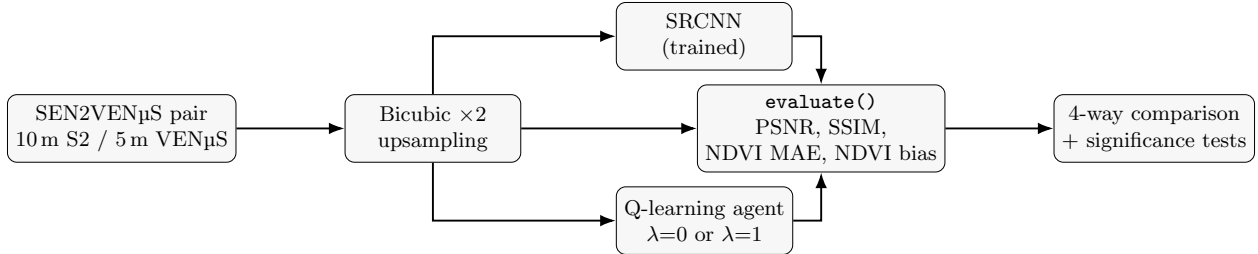


Figure 1: Graphical abstract. Every method is derived from the same bicubic starting point and scored by one shared evaluation routine, so PSNR/SSIM (pixel fidelity) and NDVI MAE/bias (radiometric fidelity) are always reported side by side for the same patches.

2.1 Dataset Description

This study uses SEN2VENµS [11] (Zenodo record 6514159), a dataset of co-registered, atmospherically corrected Sentinel-2 (10 m) and VENµS (5 m) surface reflectance patches over 29 sites worldwide and restrict this study to a single site (NARYN) to keep the radiometric comparison controlled. Table 1 summarizes its properties as verified on disk. Files are `torch.saved int16` tensors, band order B2/B3/B4/B8 (red = index 2, NIR = index 3), divided by 10000 to obtain surface reflectance; the LR input is 128×128 px and the HR target is 256×256 px, an exact $\times 2$ scale factor.

Table 1: SEN2VENµS / NARYN dataset properties.

Property	Value
Acquisition pairs	25 (2018–2020)
Total patches	3,814
LR (Sentinel-2) patch	128×128 px, 10 m, bands B2 B3 B4 B8
HR (VENµS) patch	256×256 px, 5 m, bands B2 B3 B4 B8
Scale factor	$\times 2$
Split unit	entire acquisition pair (never a random patch)
Split (pairs)	16 / 4 / 5 (train / val / test)
Split (patches)	2,478 / 442 / 894 (train / val / test)

Splitting is done at the level of whole acquisition pairs rather than individual patches: patches from the same date are strongly spatially and radiometrically correlated so a random patch-level split would leak the test set into training and inflate every metric. The resulting patch counts are imbalanced relative to the pair counts (65%/12%/23% of patches versus 64%/16%/20% of pairs) because pairs differ in patch yield by up to two orders of magnitude; this is an accepted cost of the correct split unit. A data-quality audit over every patch found no occurrence of the documented $-10,000$ no-data sentinel, but 28.3% of patches contain at least one pixel with slightly negative red-band reflectance (an L2A atmospheric-correction artifact over dark, densely vegetated or shadowed pixels); only 0.076% of all pixel-band values are affected, concentrated in the exact high-NDVI regime this project studies, so no patches are filtered rather NDVI is instead clipped to $[-1, 1]$ before every computation, which is found necessary (unclipped NDVI reaches $O(10)$ from a handful of near-zero-denominator pixels) and sufficient (clipped MAE and its unclipped counterpart agree to within 6% on a spot check). Figure 2 shows one registered LR/HR pair; visual alignment is excellent and the NDVI error field between bicubic-upsampled LR and true HR is spatially structured (concentrated along field boundaries and braided-channel edges) rather than white noise, which is what makes it plausible that a spatially aware model could learn to correct it.

2.2 Tools, Models and Algorithms

Metrics. Four per-patch numbers are computed on float reflectance by one function, `evaluate()`, so every method below is scored identically: PSNR ($10 \log_{10}(D^2/\text{MSE})$ over all four bands, $D = 1$), SSIM (Wang et al.

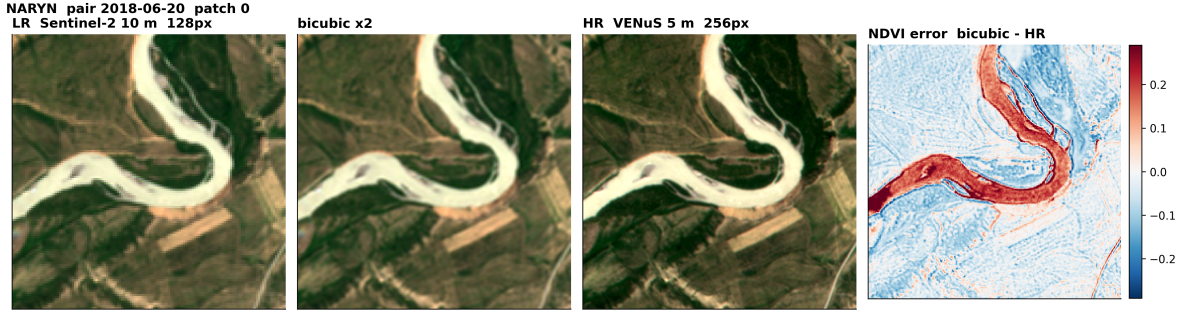


Figure 2: Registration check: LR (10 m) and HR (5 m) views of the same acquisition, confirming pixel-level alignment before any radiometric comparison is trusted.

formulation, Gaussian window $\sigma = 1.5$, averaged over bands), NDVI MAE (mean $|\text{NDVI}(\text{SR}) - \text{NDVI}(\text{HR})|$), and NDVI bias (the same difference, signed).

SRCNN. A from-scratch, 4-band SRCNN [2] (9-5-5 kernel convolutional stack) is used deliberately in place of a pretrained natural-image network: architectures such as ESRGAN [3] expect 8-bit sRGB input and would corrupt the radiometry this project measures. The network is trained to predict a residual correction to the bicubic image rather than the image itself, since reflectance is concentrated in $[0, 0.4]$ and a direct-prediction network spends most of its capacity re-learning the identity map; this residual formulation converges markedly better within the epoch budget used here (Figure 3).

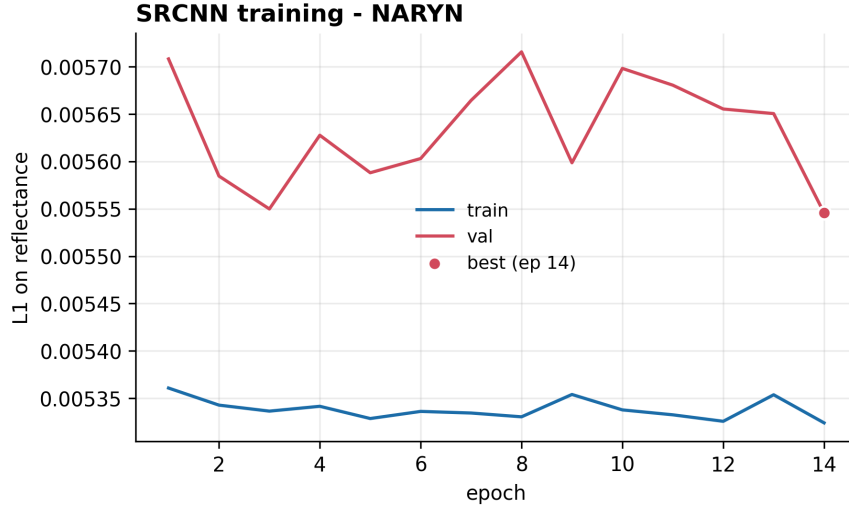


Figure 3: SRCNN training and validation loss convergence.

The super-resolution MDP and Q-learning agents. This study frames refining a super-resolved image as a Markov Decision Process (MDP) and solve it with tabular off-policy TD control (Q-learning) [9, 10]: every pixel is treated as its own agent and all pixels share one Q-table, so a single 256×256 patch supplies 65,536 transitions per step. Table 2 specifies the MDP and Table 3 the six available per-pixel actions. Actions 1 to 3 move all bands together and mainly affect PSNR; actions 4 to 5 touch only the NIR band, changing NDVI without changing the red band, the lever an NDVI-aware reward can pull that a pixel-only reward has little reason to.

Every action scales with the pixel’s own local residual (image minus a local blur) rather than moving it by a fixed amount. This is necessary for squared-error reward, a fixed perturbation δ changes error by

Table 2: The super-resolution MDP.

MDP component	Definition
State s	(local-detail bin, NDVI bin, NDVI-residual bin, step) $\rightarrow$ 240 discrete states
Action a	one of 6 per-pixel edits (Table 3)
Reward r	normalized reduction in pixel squared error, optionally plus NDVI absolute error reduction
Transition	apply the edit; the pixel moves to a new state
Update rule	$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$
Behaviour policy	ε -greedy, ε decaying linearly over training

Table 3: The six per-pixel actions.

Action	Effect	Bands
no-op	leave the pixel unchanged	–
sharpen-	gentle unsharp mask ($+0.25\times$ local detail)	all
sharpen+	aggressive unsharp mask ($+0.75\times$ local detail)	all
smooth	3×3 box blur	all
nir-sharp	unsharp mask on NIR only	NIR only
nir-smooth	box blur on NIR only	NIR only

$2\delta e + \delta^2$ and inside a state where the sign of the true error e cannot be predicted, $\mathbb{E}[e] \approx 0$, so the expected change is $+\delta^2 > 0$ strictly harmful regardless of what the agent learns. Making every action proportional to an observable local quantity is what makes a positive-expected-value policy possible at all. The two reward components live on very different numeric scales and are normalized to unit standard deviation from a 24-patch pilot under random actions (pixel term scale $5,710.7\times$, NDVI term scale $112.9\times$, a $50.6\times$ ratio between them), so that the reward weight λ is a genuine relative weight rather than an arbitrary constant. **The ablation** that this report is built around: two agents, q -psnr ($\lambda = 0$, reward is pixel error only) and q -ndvi ($\lambda = 1$, reward adds the normalized NDVI term), trained with identical seed, patches, hyperparameters and action space; the only difference between them is λ .

2.3 Performance Evaluation

All four methods (bicubic, SRCNN, q -psnr, q -ndvi) are scored on the same held-out **test** split (894 patches, 5 acquisition dates never seen during training or validation of any method). SRCNN’s validation split (442 patches, 4 dates) is used for early stopping and for the headroom analysis of Section 3.2; the training split (2,478 patches, 16 dates) is additionally re-scored in Section 3.4 as an in-sample control for the RL agents. Differences between methods on the test split are tested with the paired Wilcoxon signed-rank test (patches are matched, not independent samples) and the agreement between PSNR and NDVI MAE on which of two methods is better is tested against chance with an exact binomial test.

3 Experimental Analysis

3.1 Phase 1: does PSNR track NDVI accuracy?

Table 4 gives the headline test-split comparison between bicubic and SRCNN. SRCNN’s own gain over bicubic is small (-0.012 dB PSNR, a non-significant Wilcoxon $p = 0.40$) and is addressed directly in Section 3.2; the result this section is built around is not about SRCNN’s quality but about whether the ranking PSNR gives matches the ranking NDVI MAE gives.

Across-patch correlation. Pooling both methods ($n = 1,788$), Spearman $\rho = -0.632$ ($p \approx 3 \times 10^{-200}$) and $R^2 \leq 0.363$: PSNR explains at most 36.3% of the variance in NDVI error, leaving the majority

Table 4: Phase 1 headline comparison, test split (894 patches, 5 pairs).

Method	PSNR (dB)	SSIM	NDVI MAE	NDVI bias
Bicubic	41.7264	0.9742	0.0273	+0.0013
SRCNN	41.7145	0.9748	0.0270	−0.0027
Δ (SRCNN − bicubic)	−0.012	+0.0006	−0.0003	−0.0040

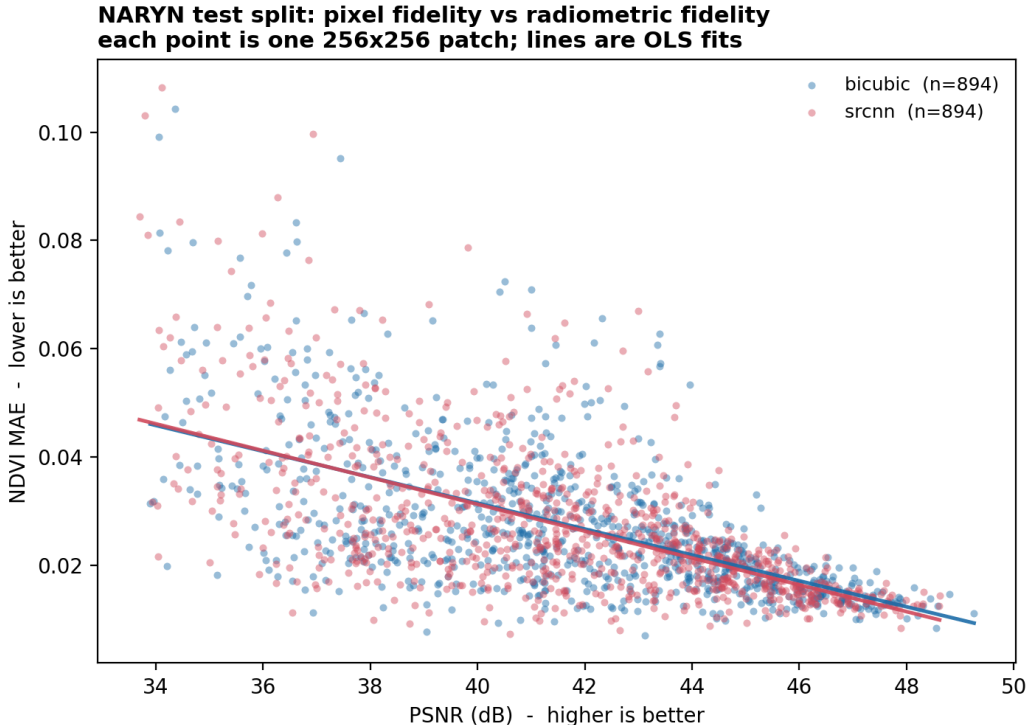


Figure 4: PSNR versus NDVI MAE, one point per test patch. The negative slope confirms the expected direction, but the spread at any fixed PSNR is large.

unexplained by pixel fidelity alone. This test is confounded by scene difficulty (an easy patch scores well on both metrics for reasons that have nothing to do with either model), so this study also run the sharper, confound-free test below. Figure 4 shows the full scatter.

Within-patch disagreement. Comparing SRCNN against bicubic on the same patch removes the scene-difficulty confound entirely. Table 5 shows that on 15.5% of test patches, PSNR and NDVI MAE disagree about which method is better; agreement is well above chance (binomial $p = 3.99 \times 10^{-103}$) but far from perfect and the disagreement is asymmetric: NDVI MAE prefers SRCNN on 10.3% of patches where PSNR prefers bicubic, versus only 5.3% the other way.

Failure cases. Restricting to the top PSNR quartile (≥ 44.47 dB, 224 patches "the model looks excellent") and taking the ten worst NDVI MAE within it surfaces patches at 44.5-45.7 dB (near the top of the entire test distribution) carrying NDVI bias between -0.014 and -0.034 : systematic browning invisible to a pixel-fidelity metric. Figure 5 shows four of these cases side by side with their NDVI error field.

3.2 How much room is there to improve on bicubic?

Before attributing SRCNN's small gain to a training defect, it is measures that how much of the bicubic residual is even recoverable. Table 6 reports validation-split L1 reflectance error for four references: plain

Table 5: Within-patch agreement between PSNR and NDVI MAE (SRCNN vs. bicubic, $n = 894$).

Outcome	Fraction of patches
Both metrics agree	84.5%
Metrics disagree	15.5%
PSNR says SRCNN better, NDVI says worse	5.3%
PSNR says SRCNN worse, NDVI says better	10.3%

bicubic; bicubic plus a per-band affine correction fitted on the training split; bicubic plus an oracle correction using the held-out date’s own mean error (information no real model has) and the trained SRCNN.

Table 6: Headroom analysis, validation split (lower is better).

Method	Validation L1	vs. bicubic
Bicubic	0.005640	–
+ per-band affine fit (train)	0.005667	+0.48% (worse)
+ oracle per-date offset (cheating)	0.005588	–0.93%
SRCNN (trained)	0.005546	–1.68%

The oracle, which is allowed to see the held-out date’s own mean error, recovers under 1% of the bicubic residual. The bicubic residual is therefore not a learnable radiometric offset; it is high-frequency detail that Sentinel-2 at 10 m physically never recorded, because SEN2VEN μ S pairs are genuinely cross-sensor (different spectral response, viewing geometry and acquisition time), unlike the synthetically downsampled pairs used in much of the SR literature reviewed in Section 1. This sets expectations for Section 3.3: no refinement policy, learned or otherwise, has much room to win here.

3.3 Phase 2: does an NDVI-aware reward change the outcome?

Table 7 gives the full four-method test-split comparison, and Table 8 the paired significance tests. Both Q-learning agents refine *bicubic*, not SRCNN, which keeps the q -psnr/ q -ndvi ablation clean (the only difference between them is λ) but means they are not competing with SRCNN on equal footing.

Table 7: Final four-method comparison, test split (894 patches).

Method	PSNR (dB)	SSIM	NDVI MAE	NDVI bias
Bicubic	41.7264	0.9742	0.0273	+0.0013
SRCNN	41.7145	0.9748	0.0270	–0.0027
q -psnr	41.6493	0.9738	0.0274	+0.0020
q -ndvi	41.4780	0.9731	0.0275	+0.0008

The reward ablation. q -ndvi attains the smallest absolute NDVI bias of all four methods (+0.0008, versus +0.0013 bicubic, –0.0027 SRCNN, +0.0020 q -psnr) but its NDVI MAE is not better than q -psnr’s (+0.00005, +0.17% relative, and this tiny difference is itself statistically significant at $p = 2.8 \times 10^{-9}$ given the paired design and large n , even though it is not practically meaningful), and it costs 0.171 dB of PSNR. So the reward term visibly reduces systematic bias without reducing average absolute error, and does so at a measurable pixel-fidelity cost.

Did the reward change behaviour? Table 9 shows the greedy policy learned by each agent. The two policies disagree on 39.2% of the 240 states, and the shift is concentrated exactly where designed: q -ndvi uses the NIR-only **smooth** and **sharpen+** actions substantially more often than q -psnr. Figure 6 tracks each reward component during training: the NDVI component improves markedly for q -ndvi ($-0.0796 \rightarrow -0.0393$ by the final 20 training batches) while the pixel component is essentially unchanged ($-0.1187 \rightarrow -0.1237$).

Table 8: Paired Wilcoxon signed-rank tests, test split ($n = 894$).

Comparison	Metric	Mean diff.	Winner’s share	p -value
q -ndvi vs. q -psnr	PSNR	−0.171 dB	12.1%	5.6×10^{-122}
q -ndvi vs. q -psnr	NDVI MAE	+0.00005	37.7%	2.8×10^{-9}
q -ndvi vs. bicubic	PSNR	−0.248 dB	5.4%	1.3×10^{-143}
q -ndvi vs. bicubic	NDVI MAE	+0.00016	30.0%	1.4×10^{-23}
SRCNN vs. bicubic	PSNR	−0.012 dB	50.0%	0.40 (n.s.)
SRCNN vs. bicubic	NDVI MAE	−0.00034	55.0%	6.7×10^{-4}

The reward mechanism is doing exactly what it was designed to do in training; it simply does not translate into a held-out MAE win.

Table 9: Greedy action frequency across all 240 states, by agent.

Action	q -psnr	q -ndvi
no-op	67.9%	60.0%
sharpen-	12.5%	12.1%
sharpen+	3.8%	6.2%
smooth	0.4%	6.7%
nir-sharp	9.2%	8.8%
nir-smooth	6.2%	6.2%
Policies disagree on 39.2% of states		

3.4 Ruling out overfitting: an in-sample check

Neither agent beats bicubic on the held-out test split. Two very different explanations are consistent with that fact: overfitting (the agent would win on its own training dates but fails to generalize) or a genuine headroom limit (no policy in this action space can win anywhere). This study distinguishes them by re-scoring q -psnr on its own **training** patches. It still loses: bicubic scores 43.3202 dB against q -psnr’s 43.2284 dB (−0.092 dB) and q -psnr beats bicubic on only 25.4% of the very patches it learned from. An agent that had merely memorized date-specific noise would be expected to win in-sample and lose out-of-sample; losing in both rules out overfitting and points to the action space itself, consistent with the headroom result of Section 3.2.

4 Conclusion

Returning to the question this project opened with: does a reward that includes NDVI accuracy preserve downstream scientific accuracy better than a reward based on pixel fidelity alone, even at equal or lower PSNR? Phase 1 answers this firmly for the underlying premise: PSNR and NDVI error decorrelate at the pixel level on real Sentinel-2/VENµS pairs, disagreeing on which of two methods is better on 15.5% of held-out patches, with the worst disagreements concentrated in patches that a PSNR-only selection criterion would call excellent. Phase 2’s reward-shaping ablation is a small, honest negative result rather than a hidden one: an NDVI-aware Q-learning agent measurably shifts its learned policy toward the NDVI objective and attains the smallest absolute bias of any method tested but this does not translate into a lower NDVI MAE, costs PSNR and neither Q-learning agent beats plain bicubic upsampling in-sample or out-of-sample. An oracle headroom analysis attributes this to the dataset itself; SEN2VENµS is a genuinely cross-sensor pairing in which most of what separates the LR input from the HR target was simply never recorded at 10 m, leaving almost no residual for any refinement policy, learned or otherwise, to recover.

Limitations. All results are measured at a single site (NARYN), a single scale factor ($\times 2$), and a single cross-sensor pairing; the headroom ceiling of Section 3.2 is a property of that specific pairing, not

of super-resolution or of reinforcement learning in general. The RL agent is deliberately simple, tabular Q-learning over six fixed, hand-designed pixel-wise actions to match the scope of a course project and to keep the reward ablation clean and interpretable; a function-approximation agent with continuous, learned actions is the natural next step, but only once a task with real headroom is chosen. A same-sensor task (e.g. $\times 4$ super-resolution from artificially downsampled VEN μ S) would have recoverable detail and is the most promising direction for future work seeking a positive RL result, at the cost of no longer testing the cross-sensor setting this project was actually designed to probe.

Code. The full, single-notebook implementation (data loading, Phase 0 sanity checks, metrics, SRCNN, the MDP and both Q-learning agents, and every table and figure in this report) is available at https://github.com/mahinur-rashid/CSE753_project.

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NARYN: high PSNR, high NDVI error - SRCNN failure cases

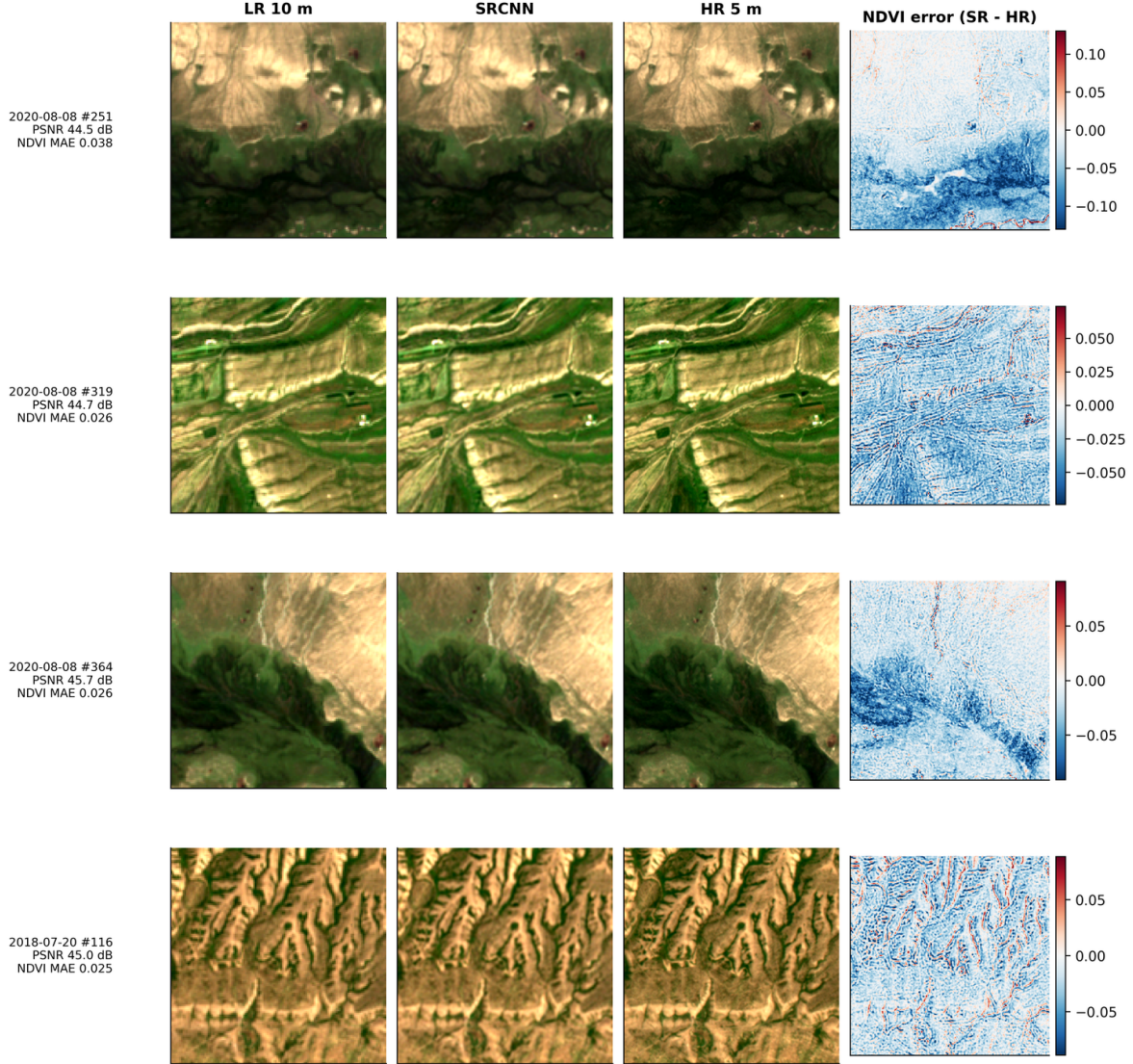


Figure 5: Failure-case gallery: patches in the top PSNR quartile with the worst NDVI error. Each row shows the LR input, the SRCNN output, the HR truth and the signed NDVI error field; the error is spatially structured, not noise and concentrated in vegetated regions.

Reward components logged separately (15-batch moving average)

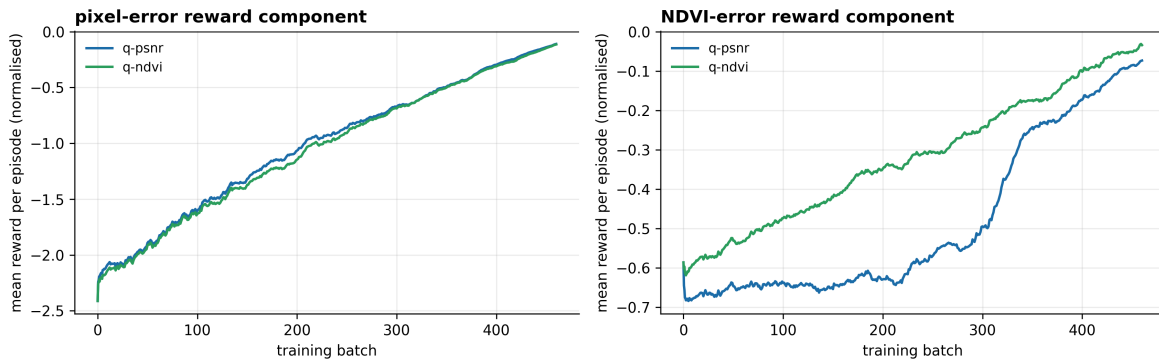


Figure 6: Training reward components (15-batch moving average). The NDVI component improves for q -ndvi while the pixel component does not move – the reward machinery works as designed during training.